Safety Manual

MPQ70165FS

MPQ70165FS Safety Manual

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Table of Contents

1 Introduction 5
1.1 Purpose 5
1.2 Scope 5
1.3 External Audits 5
1.4 Disclaimers 5
1.5 Terms and Abbreviations 6
1.6 References 6
2 Development Process for Management of Systematic Faults 7
2.1 MPSafe Product Development Lifecycle 7
2.2 External MPSafe Process Audits 7
3 Product Overview 8
4 Assumptions 10
4.1 Purpose of the SEooC 10
4.2 Assumed Safety Goal 10
4.2.1 Automotive Safety Integrity Level (ASIL) 11
4.2.2 Assumed Safe State 11
4.2.3 Fault-Tolerant Time Interval (FTTI) 11
4.2.4 Multiple-Point Fault-Detection Interval (MPFDI) 11
4.3. Assumed System Technical Safety Requirement 11
4.3.1 System_TSR_1 11
4.3.2 System_TSR_2 12
4.3.3 System_TSR_3 12
4.3.4 System_TSR_4 12
4.3.5 System_TSR_5 12
4.3.6 System_TSR_6 12
4.4 Assumed Operation Conditions 12
4.5 Assumed Hardware Environment 12
4.5.1 Assumption of External Power Supply 12
4.5.1.1 AoU_Vin Condition 12
4.5.1.2 AoU_Vin_SafeState 12
4.5.1.3 AoU_Vin_Max_Time 12
4.5.2 Assumption of Watchdog (Window Watchdog or Question and Answer) 13
4.5.3 AoU_System_PowerUp_Safety_Test 13
4.5.3.1 ALERT# Test 13

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4.5.3.2 PG# Start-up Test 14
4.5.4 Assumption of Temperature 14
4.5.5 Assumption of I2C Communication 14
4.5.6 Assumption of System Read Back 14
5 Functional Safety Concept 15
5.1 Operating States 15
5.2 External Safety Mechanisms 15
5.2.1 PG# Power-Up Test 15
5.2.2 ALERT# Power-Up Test 16
5.2.3 Write Message Readback 16
5.2.4 External Vin Monitoring 16
5.3 Internal Safety Mechanisms 16
5.3.1 I^2C CRC Communication Check 17
5.3.2 Digital Clock Monitoring 18
5.3.3 OTP CRC 18
5.3.4 Register CRC Check 19
5.3.5 Register Write Check 19
5.3.6 V_{IN} Under-Voltage Lockout (V_{IN} UVLO) 19
5.3.7 V_{cc} Under-Voltage Lockout (V_{CC} UVLO) 20
5.3.8 PG Over-Voltage (OV) Monitoring 20
5.3.9 PG Under-Voltage (UV) Monitoring 20
5.3.10 Reference Voltage Cross Check 21
5.3.11 Analog Built-In Self-Test (ABIST) 21
5.3.12 Logic Built-In Self-Test (LBIST) 22
5.3.13 VOUT Over-Voltage (OV) Monitoring 22
5.3.14 VOUT Under-Voltage (UV) Monitoring 22
5.3.15 Auxiliary Voltage Monitor 23
5.3.16 State Machine Error Detection 23

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1 Introduction

1.1 Purpose

This document is the safety manual for MPS's MPQ70165FS, a power management integrated circuit (PMIC) for advanced driver assistance system (ADAS) applications. This document describes the safety mechanisms and the actions that must be taken by the system integrator to ensure the MPQ70165FS's proper operation in a safety-relevant system.

This document contains the following:

- Overview of the safety architecture (external and internal).
- List of assumptions for the system safety requirements.
- Details of the MPQ70165FS's safety mechanisms that reduce single-point failures.

The safety-relevant documents below are not covered in this document. The summary (or short version of these documents) can be provided upon request.

- FMEDA
- Functional safety assessment report
- Certificate of MPSafe product development process

If you feel there is missing or unclear information, please contact the MPS safety team at MPSafe@monolithicpower.com. Your feedback helps us improve our documentation.

1.2 Scope

The MPQ70165FS is intended to be an element of an electronic/electrical system, as defined in ISO26262. It was developed as a safety element out of context (SEooC) to meet the assumed automotive safety integrity level (ASIL). While developing the MPQ70165FS, MPS made various assumptions, which are described in detail in the following sections.

Since this product is developed based on MPS assumptions, the actual application may differ from what is described in this document. Any deviations should be discussed with the MPS safety team to ensure that the MPQ70165FS meets actual system safety requirements. (1)

Functional safety begins at the vehicle level, and trickles down to the system and element level. It is the user or system integrator's responsibility to ensure that the MPQ70165FS is suitable for the application and complies with the appropriate application standards.

Note:

1) Here and throughout this document, "shall" indicates that this is a mandatory requirement to be fulfilled.

1.3 External Audits

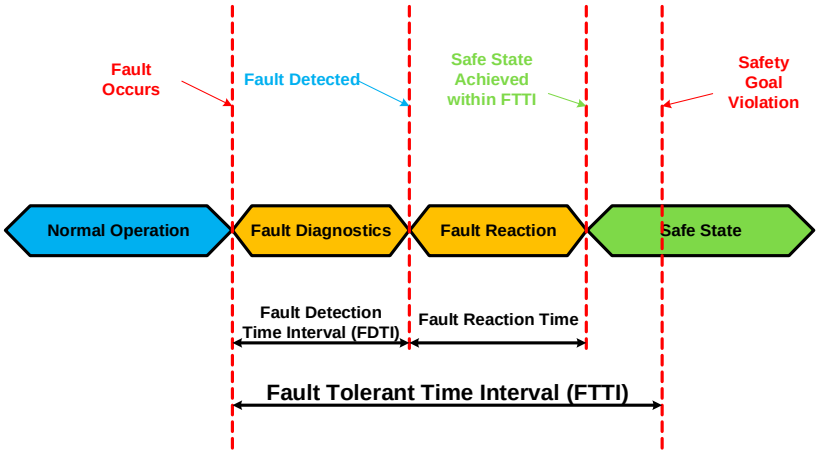
MPS has not planned to perform an external ISO26262 audit for this product. Documentation can be made available to support customer system audit and certification.

1.4 Disclaimers

The information given in this document is a hint for the implementation of the component only. It shall not be regarded as any description or warranty of a certain functionality, condition or quality of this component. The recipient of this document must verify any function described herein in the real application. MPS hereby disclaims any and all warranties and liabilities of any kind (including without limitation warranties of non-infringement of intellectual property rights of any third party) with respect to any and all information given in this document.

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1.5 Terms and Abbreviations

AoU	**Assumption of Use**
ASIL	**Automotive Safety Integrity Level** One of four levels that specifies the items or elements necessary for ISO 26262, as well as safety measures that help avoid an unreasonable residual risk. D represents the most stringent level; A is the least stringent level. (ISO 26262-1)
FDTI	**Fault Detection Time Interval** 
FMEDA	**Failure Modes, Effects, and Diagnostic Analysis**
FTTI	**Fault-Tolerant Time Interval** Time during which a fault (or faults) can be present in a system before a hazardous event occurs. (ISO 26262-1)
MPFDI	**Multiple-Point Fault-Detection Interval**
PCB	**Printed Circuit Board**
SEooC	**Safety Element Out of Context** An SEooC is a safety-related element that is not developed for a specific product, meaning it is not developed for a particular vehicle. (ISO26262-10)
TSR	**Technical Safety Requirement**

1.6 References

[1]	ISO 26262 Road Vehicles – Functional Safety *2nd edition Dec 2018* https://www.iso.org/obp
[2]	MPQ70165FS Datasheet
[3]	MPQ70165FS FMEDA
[4]	Certificate of MPSafe product development process
[5]	MPQ70165FS FTA
[6]	MPQ70165FS DFA

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2 Development Process for Management of Systematic Faults

For safety relevant product development, both systematic and random faults should be managed. The MPSafe product development process is defined to reduce the systematic faults during the safety-relevant product development cycle. This process builds on the ISO9001 MPS quality management and product reliability development process. It adjusts all the ASIL-D requirements according to ISO26262.

2.1 MPSafe Product Development Lifecycle

The MPSafe product development cycle consists of the following phases (see Figure 1):

- FSM_0: Concept
- FSM_1: Design/Implementation
- FSM_2: Sampling
- FSM_3: Verification & Validation
- FSM_4: Release for Production

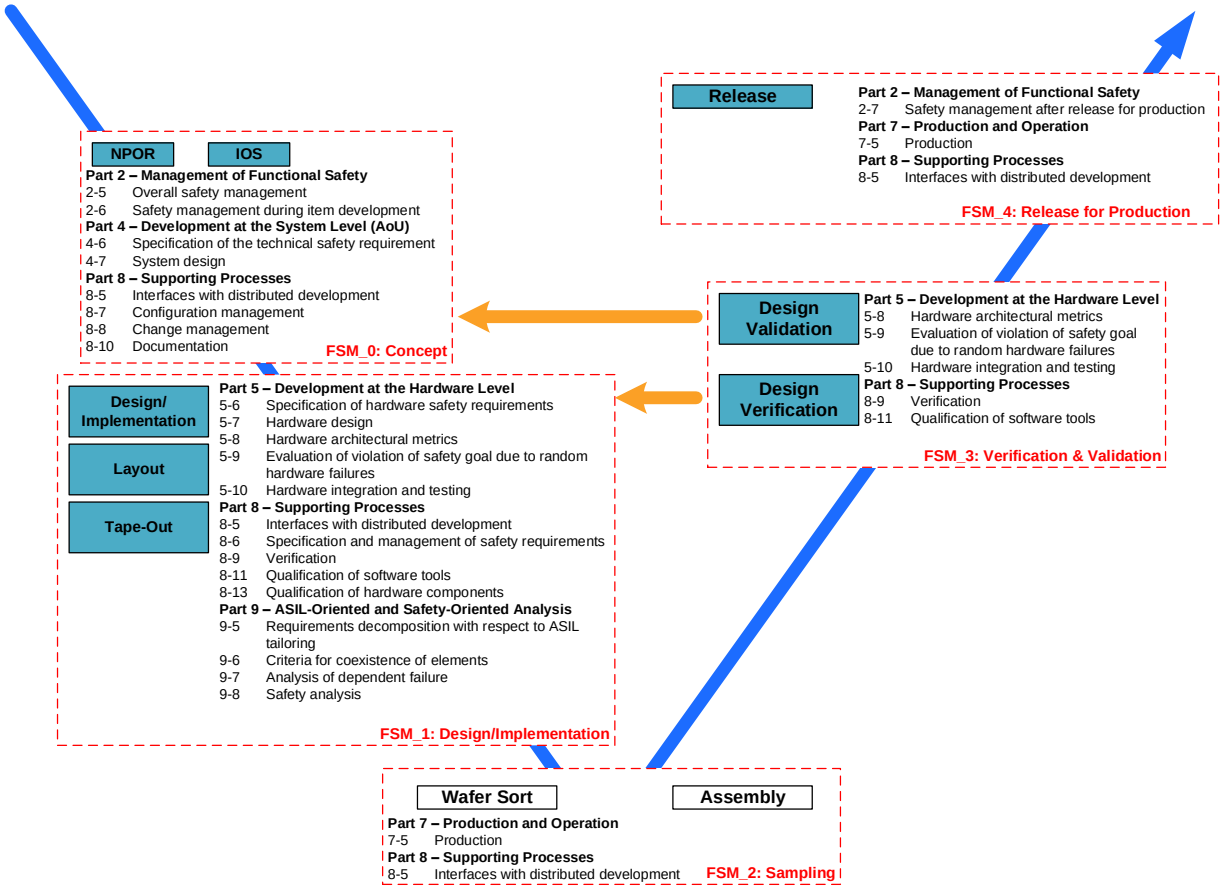


Figure 1: MPSafe Product Development Process

2.2 External MPSafe Process Audits

The MPSafe Process is certified by TUV-SUD to reduce the probability of systematic faults for the ASIL-D requirement.

The certificate is available at the MPS website. Contact your local MPS sales office for additional assistance.

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3 Product Overview

The MPQ70165FS is a Power Management IC designed to meet the power management requirements of processors used in various safety-relevant automotive systems.
Six integrated synchronous buck converters can be configured as 3 rails dual phase or 6 rails single phase converters. With dynamic voltage scaling, the output voltage of each converter can be changed during normal operation. The PMBus interface with PEC and integrated multi-page OTP memory (MOTP) allows for a high degree of configurability.

To prevent overshoot during start-up, soft-start functionality is featured on each converter. In addition, to allow proper discharge during shut-down, soft-stop functionality is featured on each converter. The slew rate for both functions is configurable through the digital registers.

A configurable windowed or Q&A watchdog and dedicated WDI pin, external voltage monitor, and sequence output pin are all featured.

2MHz fixed frequency peak current mode control is used to regulate output voltage, offers fast transient performance, and allows for a large reduction in external inductor and capacitor values. Frequency Spread Spectrum reduces EMI noises. A full protection suite is included and features under voltage lock-out (UVLO), under voltage protection (UVP), over voltage protection (OVP) and thermal shutdown.

Safety mechanisms such as built-in self-test (BIST) provide high diagnostic coverage, which help the system achieve its target ASIL rating. It is developed under MPS' advanced MPSafe™ Functional Safety Product Development process, which has been independently certified to meet ISO26262 guidelines.

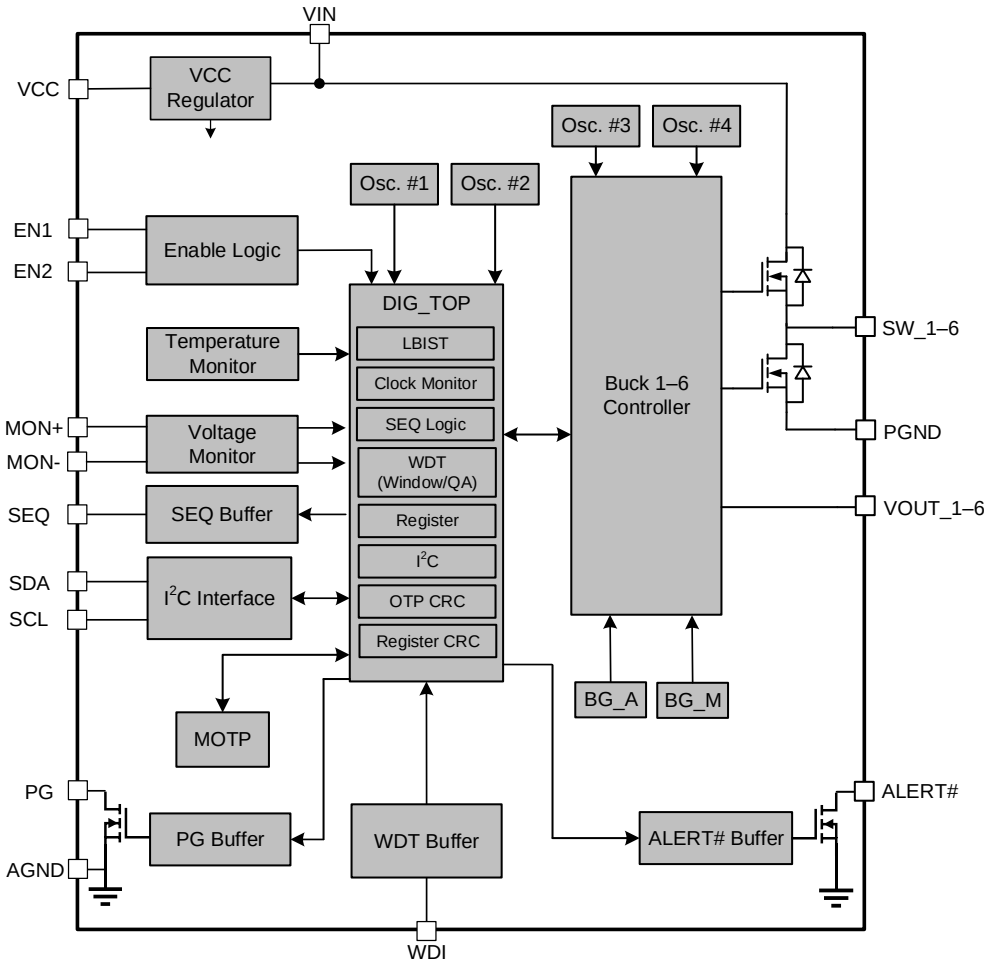


Figure 2: Functional Block Diagram

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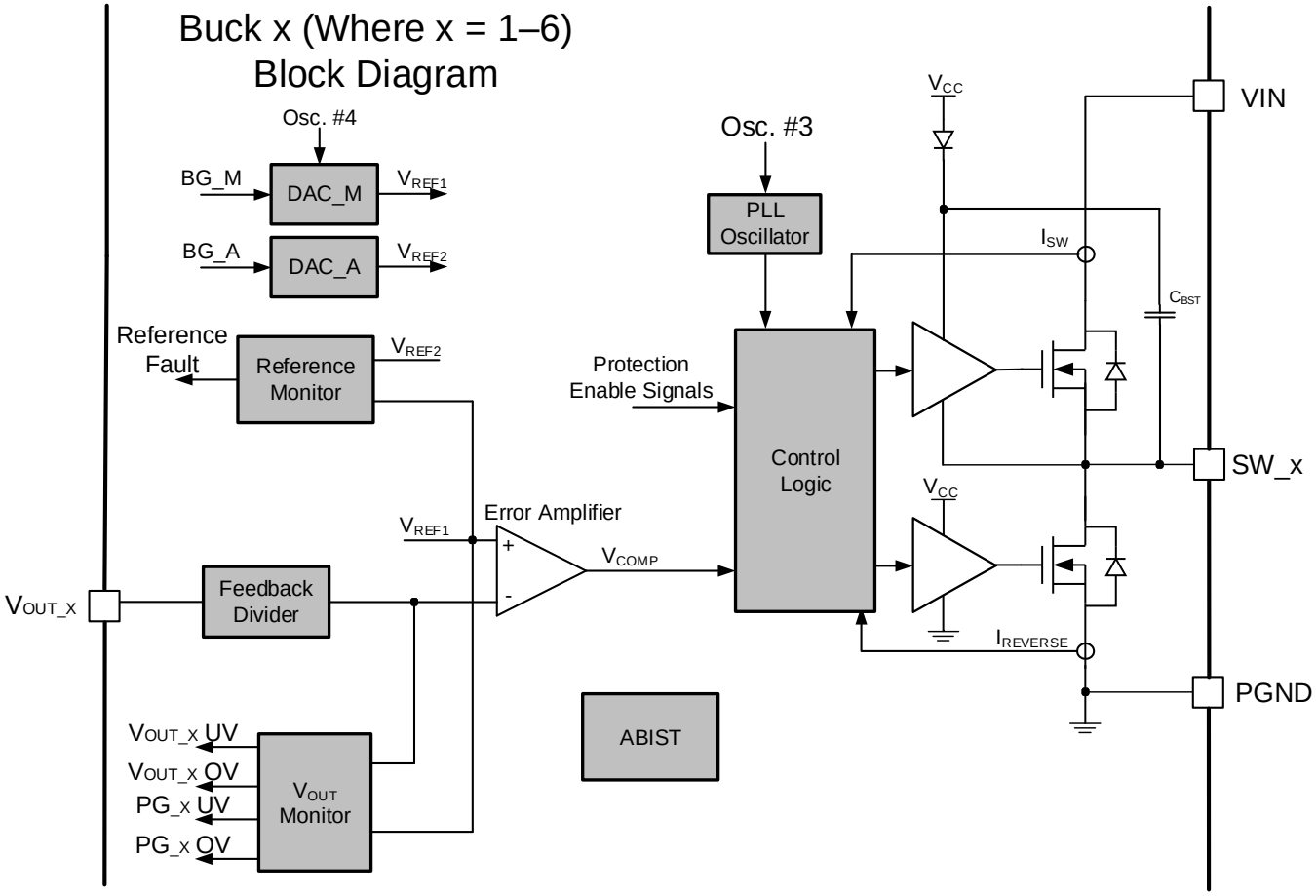


Figure 3: Buck X (Where X = 1–6) Block Diagram

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4 Assumptions

The MPQ70165FS is developed as a safety element out of context (SEooC) following the guidelines for SEooC hardware component development according to ISO26262-10, Clause 9. The according assumptions for system-level technical safety requirements are specified and described in this section.

4.1 Purpose of the SEooC

The MPQ70165FS is intended to be used as hardware in an electronic control unit according to ISO26262-10, Clause 4.2. Figure 4 shows the MPQ70165FS's assumed purpose.

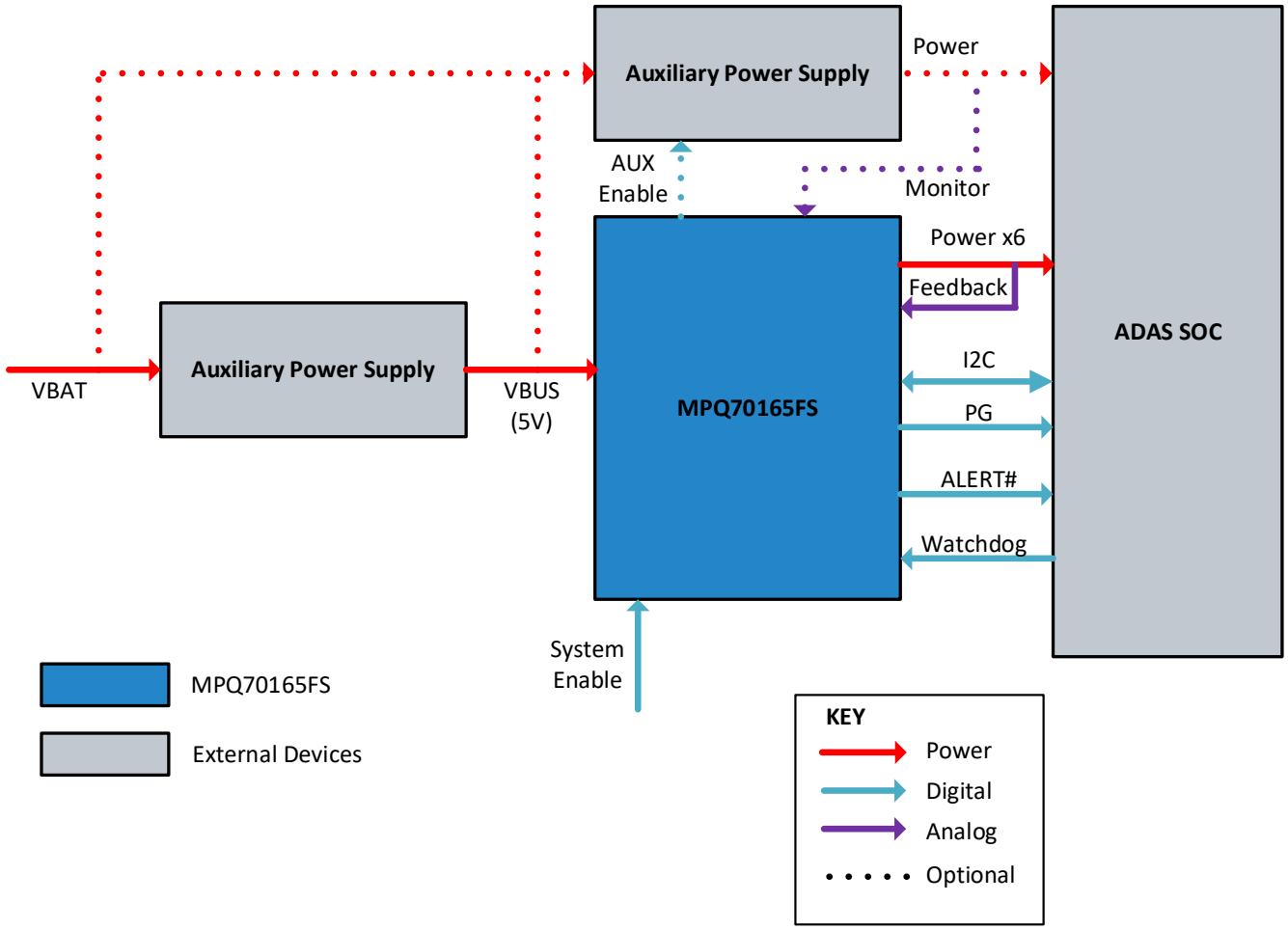


Figure 4: Typical Application Diagram

The MPQ70165FS:

- Is a highly flexible power solution for ADAS SoCs.
- Integrates six high-frequency, synchronous, rectified step-down converters.
- Is implemented with a PMBus and MOTP memory that allows for significant configurations to meet the requirements of the target SoC.

4.2 Assumed Safety Goal

The safety goals are defined at the vehicle level. As a result, the MPQ70165FS does not assume any safety goals. However, the device's failure modes may violate the safety goals at the system level. These failure modes are listed below:

- Rail(s) over-voltage (OV) failure.

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- Rail(s) under-voltage (UV) failure.

4.2.1 Automotive Safety Integrity Level (ASIL)

The MPQ70165FS is compliant up to ASIL-B requirements for the safety goals and assumptions. For more details on the related failure modes, see Section 4.2 above

4.2.2 Assumed Safe State

The assumed default safety state in this document is when the device is off. As a result, the following failure modes are not considered a safety-relevant failure:

- Any failure modes that cause the device to unexpectedly shut down

4.2.3 Fault-Tolerant Time Interval (FTTI)

The MPQ70165FS does not assume a fault-tolerant time interval (FTTI) since it is application-dependent. However, the actual fault detection time interval (FDTI) is assumed to be 5ms, which must be shorter than the actual application's FTTI. Figure 4 shows the FTTI.

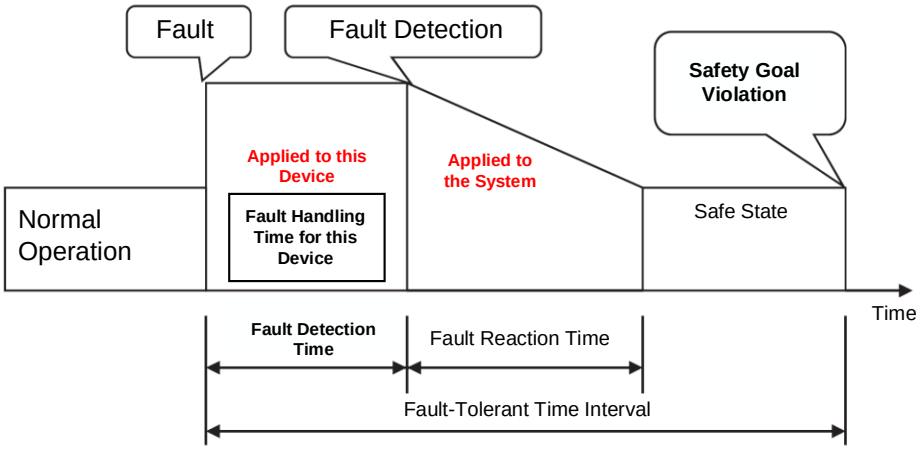


Figure 5: Fault-Tolerant Time Interval

4.2.4 Multiple-Point Fault-Detection Interval (MPFDI)

All safety mechanisms used by the application shall be tested at least once per multiple-point fault-detection interval (MPFDI) to prevent multiple-point faults from becoming latent.

The MPFDI is assumed to be one power cycle. As a result, the safety mechanisms shall be tested either at MPQ70165FS start-up or shutdown. It is recommended to run the test(s) at start-up. One power cycle shall be performed every 72 hours or sooner. If there are any deviations, the FMEDA data may be experiencing interference. Contact MPS for further assistance.

4.3. Assumed System Technical Safety Requirement

This section assumes that the safety requirements are derived from the system to the MPQ70165FS. The MPQ70165FS IC safety mechanisms are built around those system safety requirements. If there are any misalignments between the actual system safety requirements and the assumed system safety requirements in this section, contact MPS for additional assistance

4.3.1 System_TSR_1

The MPQ70165FS shall detect PG UV failures.

- Fault detection time interval: 5ms
- Fault reaction:
  - Output rails shut off (QM)

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- Pull the PG# low (ASIL B)
- Pull the ALERT# pin low (ASIL B)
- Set the error register STATUS_VOUT (0x7A), bit [1] (e.g. PG_UV) (QM)

4.3.2 System_TSR_2

The MPQ70165FS shall detect PG OV failures.

- Fault detection time interval: 5ms
- Fault reaction:
  - Output rails shut off (QM)
  - Pull the PG# low (ASIL B)
  - Pull the ALERT# pin low (ASIL B)
  - Set the error register STATUS_VOUT (0x7A), bit [2] (PG_OV) (QM)

4.3.3 System_TSR_3

The probabilistic metric for hardware failure (PMHF) shall be less than 10 failure in time (FIT) for OV and UV failures.

4.3.4 System_TSR_4

The MPQ70165FS achieves a single-point fault metric (SPFM) that shall be equal to or exceed 90% for both OV and UV failures.

4.3.5 System_TSR_5

The MPQ70165FS achieves a latent fault metric (LFM) that shall be equal to (or exceed) 60% for both OV and UV failures.

4.3.6 System_TSR_6

The absence or failure of a safety mechanism implemented by the MPQ70165FS shall be detected within an MPFDI (one power cycle). See Section 4.2.4 on page 11 for more details.

4.4 Assumed Operation Conditions

It is assumed that the operation conditions listed in the MPQ70165FS datasheet are not exceeded.

4.5 Assumed Hardware Environment

At the system level, there are safety measures that are assumed or recommended to be implemented by an external hardware device other than the MPQ70165FS.

4.5.1 Assumption of External Power Supply

4.5.1.1 AoU_Vin Condition

The system should ensure that the MPQ70165FS is supplied an input voltage (V_{IN}) within the operating conditions listed in the datasheet.

If the V_{IN} applied to the MPQ70165FS is out of the operating range, the MPQ70165FS's safety mechanisms cannot be guaranteed.

4.5.1.2 AoU_Vin_SafeState

The system shall transition to or maintain its safe state if V_{IN} is out of its range, which is defined in the MPQ70165FS datasheet.

4.5.1.3 AoU_Vin_Max_Time

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The system shall not use the maximum operating voltage for V_{IN} defined in the MPQ70165FS datasheet as its nominal operating voltage.

This condition may increase the MPQ70165FS's failure rate.

A defined system safe state shall be triggered if an input failure is detected.

4.5.2 Assumption of Watchdog (Window Watchdog or Question and Answer)

The MPQ70165FS uses a watchdog function to monitor the integrity of the microcontroller (MCU). If the MCU experiences an internal failure, MPQ70170FS shall detect those failures via the watchdog interface.

4.5.3 AoU_System_PowerUp_Safety_Test

It is recommended to test the system's safety path at start-up, even though these tests could be executed at any time within the multiple-point fault interval according to ISO 26262.

4.5.3.1 ALERT# Test

The safety path from MPQ70165FS' point of view shall be the connection between ALERT# and system. The safety path shall be tested every drive cycle to ensure that the system can detect the correct state of ALERT#. The system shall set the dedicated ALERT# register (ALERT_PG_FORCE (0xC0) bit 1) to pull it low at start-up. In case the ALERT# is not being properly pulled low, the system shall maintain at its safe state. No normal operation is allowed.

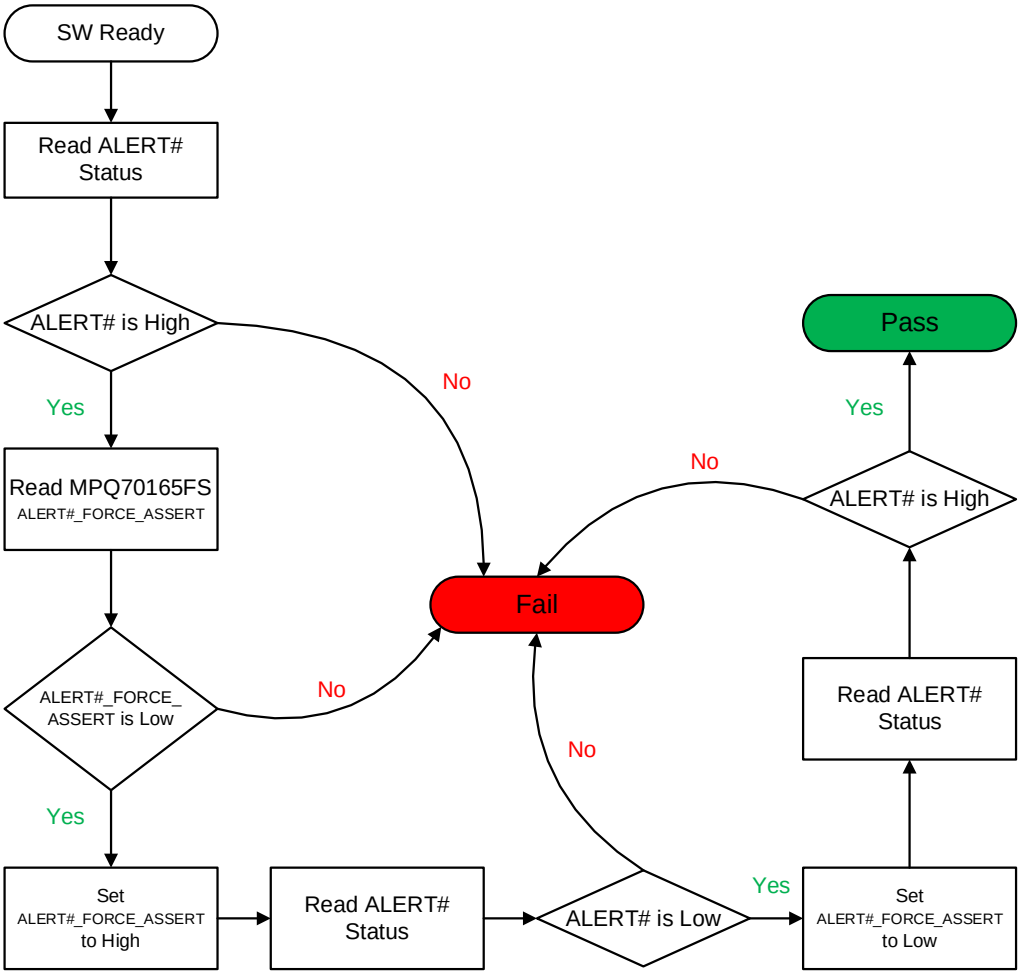


Figure 8: System ALERT# Test

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4.5.3.2 PG# Start-up Test

The safety path from MPQ70165FS' point of view shall be the connection between PG# and system. The safety path shall be tested every drive cycle to ensure that the system can detect the correct state of PG#. Therefore, system shall set the dedicated PG# register (ALERT_PG_FORCE (0xC0) bit 0) to pull PG# low at start-up. In case the PG# is not being properly pulled low, the system shall maintain at its safe state, and no normal operation is allowed.

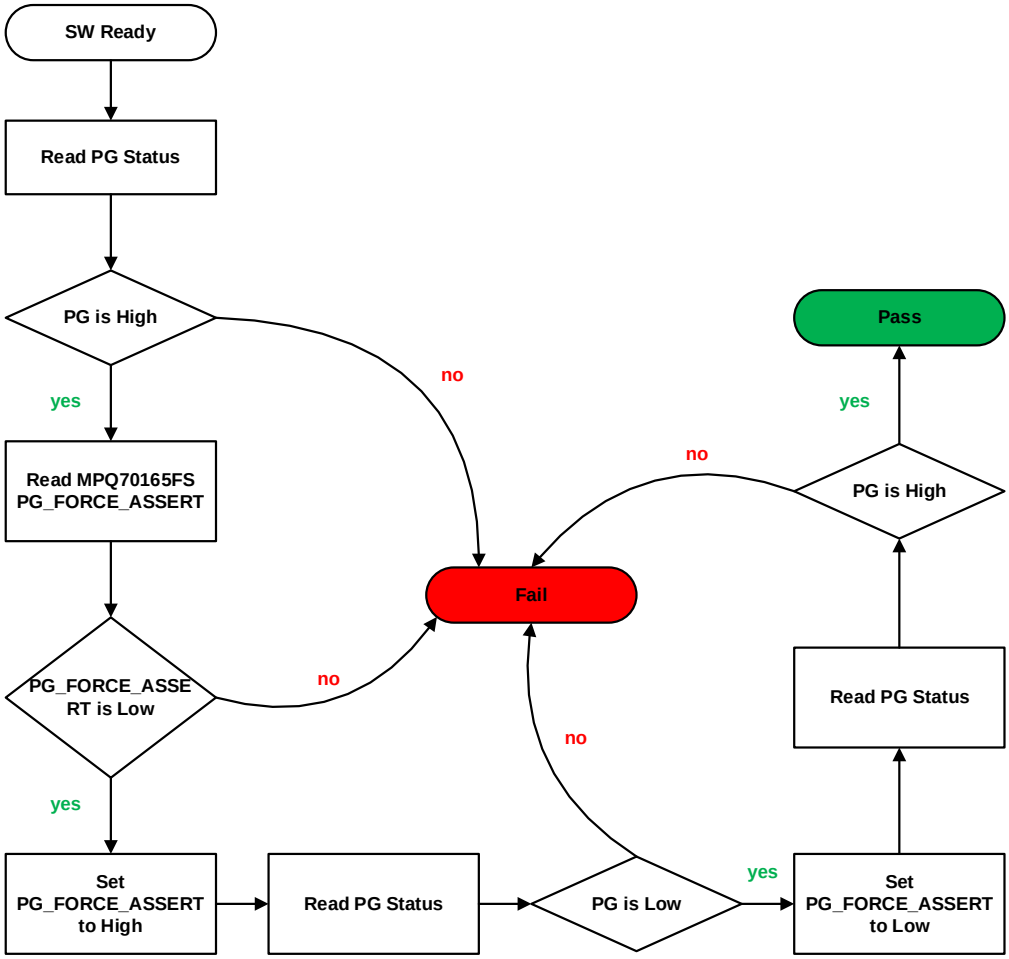


Figure 9: System PG# Test

4.5.4 Assumption of Temperature

The MPQ70165FS shall operate at the nominal junction temperature, which is below 150°C.

4.5.5 Assumption of I2C Communication

The I^2C message shall contain a CRC-8, which is represented by the polynomial $C(x) = x^8 + x^2 + x + 1$.

4.5.6 Assumption of System Read Back

A register write check shall be implemented by system to detect if written data is the same as the command.

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5 Functional Safety Concept

5.1 Operating States

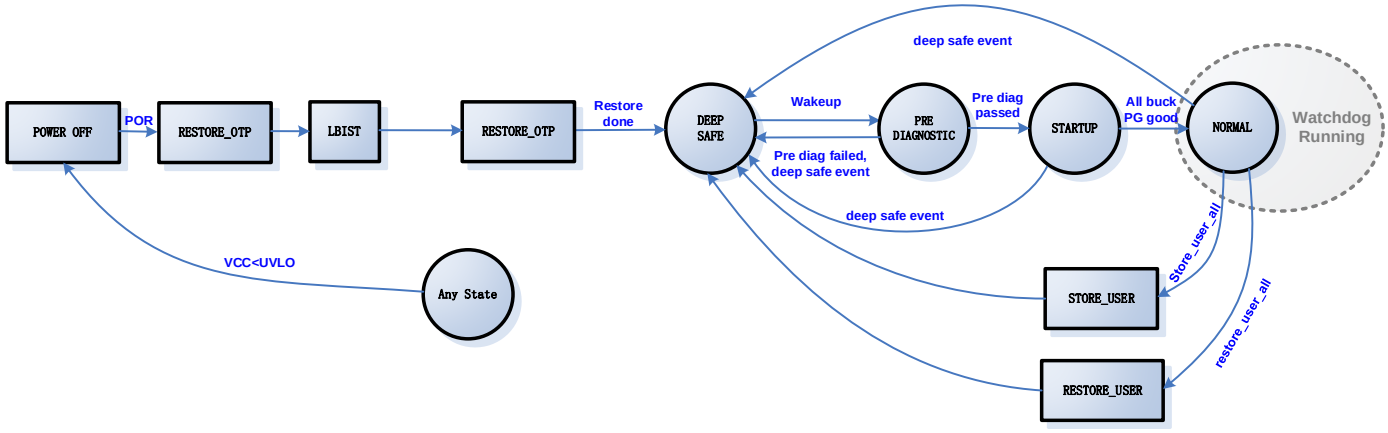


Figure 9: State Diagram

5.2 External Safety Mechanisms

The external safety mechanisms diagnostic coverage, and associated failure modes are described in detail in MPQ70165FS FMEDA. The system integrator must ensure diagnostic coverage at the system level. The safety mechanisms in this section shall be implemented by the system to improve PMHF and SPFM. Each safety mechanism is specified according to the format shown below.

ID	The identifier from FMEDA.
Description	A description.
Error Type	The error that detected by this safety mechanism. Moreover, it shall be indicated whether the detected error is a single-point fault or latent fault.
Error Signaling	The system's safe state/warning concept.
Fault-Handling Time Interval	The time required by the safety mechanism to detect the error at the system level.
Self-Test	The MPQ79700FS power-up testing to ensure the functionality of this mechanism (if needed).

5.2.1 PG# Power-Up Test

ID	SM_18
Description	The system shall toggle the PG# at power up to ensure the PG# function is working properly in case of power good failures.
Error Type	PG# stuck at failure (latent fault).
Error Signaling	The system shall remain in its safe state. Normal operation is not allowed.
Fault-Handling Time Interval	One power cycle.
Self-Test	N/A

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5.2.2 ALERT# Power-Up Test

ID	SM_19
Description	The system shall toggle the ALERT# at power up to ensure the ALERT# function is working properly in case of relevant failures.
Error Type	ALERT# stuck at failure (latent fault).
Error Signaling	The system shall remain in its safe state. Normal operation is not allowed.
Fault-Handling Time Interval	One power cycle.
Self-Test	N/A

5.2.3 Write Message Readback

ID	SM_20
Description	The SoC or microcontroller (MCU) shall read back the written value from the register and check this value against the commanded value.
Error Type	An incorrect register value has been written (single-point fault/latent fault).
Error Signaling	The system shall define a proper fault reaction.
Fault Handling Time Interval	To be defined by the system within the FTTI.
Self-Test	N/A

5.2.4 External Vin Monitoring

ID	SM_16
Description	The system shall monitor the Vin supply voltage (see Section 4.5.1 on page 12).
Error Type	MPQ70165FS power supply out of range failure (single-point fault/latent fault).
Error Signaling	The system shall define the proper state / reaction at the system level in case of Vin failure. The defined safe state / reaction shall be independent to Vin power source to avoid common cause failure.
Fault-Handling Time Interval	To be defined by the system integrator.
Self-Test	N/A

5.3 Internal Safety Mechanisms

The safety mechanisms in this section are defined via FMEDA to improve the MPQ70165FS's random hardware failure. Each safety mechanism is specified according to the format shown below.

ID	The identifier from FMEDA.
Description	A description.
Error Type	The error that is detected by this safety mechanism. Moreover, it shall be indicated whether the detected error is a single-point fault or latent fault.
Error Signaling	The error signaling types are classified as follows: • List of all the expected fault reactions

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Fault Handling Time Interval	The time required by the safety mechanism to handle the error.
Assumptions of Use (AoU)	The action taken by the integrator to initialize this safety mechanism.
Self-Test	The built-in self-test that detects the safety mechanism's failure/unavailability.

5.3.1 I²C CRC Communication Check

ID	SM_1
Description	A CRC check shall be implemented to detect at least 2 bits of failure in I²C communication. The reaction shall be triggered when a 1-bit error occurs. • Number of bits to check: 32 bits • Polynomial: 0x83 • Hamming Distance: 3
Error Type	I²C communication failure (Single-point fault/Latent fault)
Error Signaling	• Pull ALERT# low (can be configured via ALERT_MAPPING (0xC4), bit [2]) (ASIL B) • Set the error register STATUS_CML (0x7E), bit [5] (QM) • Pull PG low (after a certain delay time, PG goes back to high, as configured via PG_MAPPING (0xC6), bit [2]; the PG delay time can be set via PG_ALERT_DELAY, bits [1:0]) (ASIL B)
Fault Handling Time Interval	50µs
Assumptions of Use (AoU)	The system shall use the same CRC polynomial. In case of I2C communication failure, the system may not be able to read the proper error register. However, the ALERT# and PG# still triggered. As a result, the system shall define the proper reaction for the error that only indicated by the ALERT# and PG# when the type of error is uncertain.
Self-Test	LBIST(SM_12)

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5.3.2 Digital Clock Monitoring

ID	SM_2
Description	Two independent digital clocks are implemented as CLK1 and CLK2. CLK1 and CLK2 shall start counting at the same time. When the counter in CLK1 reaches 1000 cycles, count number in CLK2 shall be checked. If the difference between the CLK1 and CLK2 values is lower/higher than the threshold (±10%/20%/30% configurable), CLK failure shall be reported.
Error Type	Digital clock failure (Single-point fault)
Error Signaling	• Pull ALERT# low (ASIL B) • Pull PG low (ASIL B) • Output rails shut off (QM) • Set the error register STATUS_FREQ 0xE7 bit [0] (QM)
Fault Handling Time Interval	3 ms
Assumptions of Use (AoU)	The digital functions cannot be assured including set the proper error registers and shut down output rails while the digital clk failure occurs. However, the ALERT# and PG# shall be assured since it is triggered by the combination logic and analog function (digital clk is not needed). As a result, the system shall define the proper reaction for the error that only indicated by the ALERT# and PG# when the type of error is uncertain.
Self-Test	LBIST(SM_12)

5.3.3 OTP CRC

ID	SM_3
Description	The cyclic redundancy check (CRC) shall be implemented to detect at least 2 bits of error. The reaction shall be triggered when a 1-bit error occurs. • Polynomial: 0xC002 • Hamming Distance: 3
Error Type	Up to 2 bits of OTP failure (Single-point fault/Latent fault)
Error Signaling	• Pull ALERT# low (can be configured via ALERT_MAPPING (0xC4), bit [4]) (ASIL B) • Set the error register MFR_OTP_MEM_STATUS (0xDB), bit [5] (QM) • Output (by default) is off (QM) • PG is pulled low (by default) (ASIL B)
Fault Handling Time Interval	One power cycle
Assumptions of Use (AoU)	The error handling mechanisms shall be defined at the system level regarding the OTP CRC failure.
Self-Test	LBIST(SM_12)

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5.3.4 Register CRC Check

ID	SM_4
Description	A CRC check shall be implemented to detect at least 2 bits of failure for the registers. The reaction shall be triggered when a 1-bit error occurs. • Polynomial: 0xC002 • Hamming Distance: 3 This CRC check can be configured as: • By demand (default): MFR_OTP_MEM_CHK (0xDA), bit [0] transfers from 0 to 1; based on bit [2] = 0 • Continue: set MFR_OTP_MEM_CHK (0xDA), bit [2] = 1, bit [0] = 1
Error Type	Unintended change to the register value (single-point failure/latent fault)
Error Signaling	• Pull ALERT# low (can be configured via ALERT_MAPPING (0xC4), bit [0]) (ASIL B) • Set the error register MFR_OTP_MEM_STATUS (0xDB), bit [2] (QM) • Pull PG low (can be configured via PG_MAPPING (0xC6), bit [0]) (ASIL B) • Output rails shut off (QM)
Fault Handling Time Interval	3 ms
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case of register CRC failure gets reported.
Self-Test	LBIST(SM_12)

5.3.5 Register Write Check

ID	SM_5
Description	A register write check shall be implemented to detect whether the written data is the same as the intended written value.
Error Type	Register write failure (single-point failure/latent fault)
Error Signaling	• Pull ALERT# low (ASIL B) • Set the error register STATUS_FREQ 0xE7, bit [2] (QM)
Fault Handling Time Interval	50 us
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case of register write check failure gets reported.
Self-Test	LBIST(SM_12)

5.3.6 V_{IN} Under-Voltage Lockout (V_{IN} UVLO)

ID	SM_6
Description	The MPQ70165FS shall shut off if V_{IN} falls below its under-voltage lockout (UVLO) threshold. Therefore, no unintended functionalities shall occur.
Error Type	V_{IN} UVLO failure
Error Signaling	• MPQ70165FS shut off (QM) • Pull low PG# (ASIL B)

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Fault Handling Time Interval	150 µs
Assumptions of Use (AoU)	The VIN input shall be within the operating range. The system shall define the proper state / reaction in case of MPQ70165FS shuts off. See Section 4.5.1 on page 12 for more details.
Self-Test	N/A

5.3.7 V_{cc} Under-Voltage Lockout (V_{CC} UVLO)

ID	SM_7
Description	The MPQ70165FS shall shut off if V_{CC} falls below its UVLO threshold. Therefore, no unintended functionalities shall occur.
Error Type	V_{CC} UVLO failure.
Error Signaling	• The MPQ70165FS shuts off (QM) • Pull low PG# (ASIL B)
Fault Handling Time Interval	50µs
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case of MPQ70165FS shuts off
Self-Test	N/A

5.3.8 PG Over-Voltage (OV) Monitoring

ID	SM_8
Description	Monitor the output voltages on SW1, SW2, SW3, SW4, SW5, and SW6 against the over-voltage (OV) thresholds defined via PG_CONFIG (0xCA), bit [0].
Error Type	OV at buck 1, buck 2, buck 3, buck 4, buck 5, or buck 6 (single-point failure)
Error Signaling	• Set the error register STATUS_VOUT (0x7A), bit [2] (QM) • Pull low PG# (ASIL B) • Output rails shut off (QM): ○ SAFETY_CONFIG 0xC2 bit 1=0: output rails remain the same ○ SAFETY_CONFIG 0xC2 bit 1=1: output rails shut off
Fault Handling Time Interval	50 µs
Assumptions of Use (AoU)	The integrator sets this OV detection threshold via register PG_CONFIG (0xCA), bit [0]. The threshold is marked as a critical safety calibration value. Any change to this value should be handled appropriately. The system shall define the proper state / reaction in case of this failure gets reported.
Self-Test	ABIST (SM_11).

5.3.9 PG Under-Voltage (UV) Monitoring

ID	SM_9
Description	Monitors the output voltages on SW1, SW2, SW3, SW4, SW5, and SW6 against the under-voltage (UV) thresholds defined via PG_CONFIG (0xCA), bit [0].

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Error Type	UV at buck 1, buck 2, buck 3, buck 4, buck 5, or buck 6 (single-point failure)
Error Signaling	• Set the error register STATUS_VOUT (0x7A), bit [1] (QM) • Pull low PG# (ASIL B) • Output rails shut off (QM): ○ SAFETY_CONFIG 0xC2 bit 1=0: output rails remain the same ○ SAFETY_CONFIG 0xC2 bit 1=1: output rails shut off
Fault Handling Time Interval	50 µs
Assumptions of Use (AoU)	The integrator sets the UV detection threshold via PG_CONFIG (0xCA), bit [0]. The threshold is marked as a critical safety calibration value. Any change to this value should be handled appropriately. The system shall define the proper state / reaction in case of PG UV.
Self-Test	ABIST (SM_11).

5.3.10 Reference Voltage Cross Check

ID	SM_10
Description	The primary reference voltage (V_{REF1}) shall be compared to the secondary reference voltage (V_{REF2}). V_{REF1} and V_{REF2} shall come from different bandgaps.
Error Type	Reference voltage failures (single-point failure)
Error Signaling	• Pull the ALERT# low via register ALERT_MAPPING (0xC4), bit [1] (ASIL B) • Set the error register STATUS_ MFR_SPECIFIC (0x80), bit [2] (QM) • Pull the PG# low via register PG_MAPPING (0xC6), bit [1] (ASIL B) • Output rails shut off (QM): ○ PG_MAPPING 0xC6 bit 1=0 output rails shut off ○ PG_MAPPING 0xC6 bit 1=1 output rails remain the same
Fault Handling Time Interval	20 µs
Assumptions of Use (AoU)	This failure could be a common failure for OV and UV monitoring. As a result, the system shall define the associated reaction for reference voltage cross check error.
Self-Test	ABIST (SM_14).

5.3.11 Analog Built-In Self-Test (ABIST)

ID	SM_11
Description	An analog built-in self-test (ABIST) shall be performed during start-up. The defined test values check the functionality of the comparators as shown below. Fault injection test shall be performed on the following comparators during power up: • DAC reference monitoring comparators 1–12 • MON+/MON- monitoring comparators 1–2 • PG voltage comparators 1–6
Error Type	Analog-related safety mechanism failures (latent fault).
Error Signaling	• Stuck in deep safe state during power-up (output rails by default low), and re-run ABIST until ABIST passes (Enter deep safe state via register PG_MAPPING 0xC6 bit 0)

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	• Set error register: STATUS_ MFR_SPECIFIC 0x80 bit 1 (QM) • Pull low ALERT# pin via register ALERT_MAPPING 0xC4 bit 0 (ASIL B) • PG is pulled low (by default) (ASIL B)
Fault Handling Time Interval	One power cycle
Assumptions of Use (AoU)	In case of an ABIST failure, the system shall remain in the defined safe state. Normal operation shall not be allowed.
Self-Test	N/A

5.3.12 Logic Built-In Self-Test (LBIST)

ID	SM_12
Description	Testing all digital logic during start-up. When LBIST fails, MPQ70165FS enters the DEEP SAFE state. The diagnostic coverage shall be greater than 80%.
Error Type	Digital logic gate failures (latent fault).
Error Signaling	• Output rails stuck at low. POR fails to proceed. • Set the error register STATUS_MFR_SPECIFIC (0x80), bit [3] (QM) • PG is pulled low (by default) (ASIL B)
Fault Handling Time Interval	One power cycle
Assumptions of Use (AoU)	In case of an LBIST failure, the system shall remain in the defined safe state. Normal operation shall not be allowed.
Self-Test	N/A

5.3.13 VOUT Over-Voltage (OV) Monitoring

ID	SM_13
Description	Monitors the voltage outputs on SW1, SW2, SW3, SW4, SW5, and SW6 against the over-voltage (OV) thresholds (+10%).
Error Type	OV at buck 1, buck 2, buck 3, buck 4, buck 5, or buck 6 (single-point failure)
Error Signaling	• Set the error register STATUS_VOUT (0x7A), bit [7] (QM) • Pull low PG# (ASIL B) • Output rails shut off (QM) • Pull low ALERT# (ASIL B)
Fault Handling Time Interval	20 µs
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case of this failure gets reported.
Self-Test	N/A

5.3.14 VOUT Under-Voltage (UV) Monitoring

ID	SM_14

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Description	Monitors the voltage outputs on SW1, SW2, SW3, SW4, SW5, and SW6 against the under-voltage (UV) thresholds (-10%).
Error Type	UV at buck 1, buck 2, buck 3, buck 4, buck 5, or buck 6 (single-point failure)
Error Signaling	• Set the error register STATUS_VOUT (0x7A), bit [4] (QM) • Pull low PG# (ASIL B) • Output rails shut off (QM) • Pull low ALERT# (ASIL B)
Fault Handling Time Interval	20 µs
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case of this failure gets reported.
Self-Test	N/A

5.3.15 Auxiliary Voltage Monitor

ID	SM_15
Description	There is an external voltage monitoring channel that is independent from the output rails with a separated reference voltage, over-voltage (OV) and under-voltage (UV) threshold registers, and comparators. Over voltage threshold register: VMON_HIGH_THR 0xD2 Under voltage threshold register: VMON_LOW_THR 0xD1 The fault reaction shall be triggered when OV/UV failures happen.
Error Type	OV/UV at buck 1, buck 2, buck 3, buck 4, buck 5, or buck 6 (single-point failure)
Error Signaling	• Set error register: ○ over-voltage: STATUS_ MFR_SPECIFIC 0x80 bit 6 (QM) ○ under-voltage: STATUS_ MFR_SPECIFIC 0x80 bit 5 (QM) • Pull low PG# (ASIL B) • Output rails shut off (QM): ○ SAFETY_CONFIG 0xC2 bit 1=0 output rails remain the same ○ SAFETY_CONFIG 0xC2 bit 1=1 output rails shut off
Fault Handling Time Interval	50 µs
Assumptions of Use (AoU)	The system shall set the OV/UV detection threshold. The threshold is marked as a critical safety calibration value. Any change to this value shall be handled appropriately.
Self-Test	ABIST (SM_11)

5.3.16 State Machine Error Detection

ID	SM_22
Description	Single bit error in the state machine shall be detected.
Error Type	Unintended state change/stuck-at (latent fault).
Error Signaling	• Set error register: STATUS_FSM 0xE6 bit 6 (QM) • Pull low PG# (ASIL B)

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	• Pull low ALERT# (ASIL B) • Output rails shut off (QM)
Fault Handling Time Interval	50 µs
Assumptions of Use (AoU)	The system shall define the proper state / reaction in case state machine error gets reported.
Self-Test	LBIST(SM_12)

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